

Exploratory Data Analysis of Customer Shopping Data Using Python

Task 3 Documentation & Analytical Evaluation

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Qualification:	B.Tech – ECE	Domain:	Data Science Internship

1. EXECUTIVE SUMMARY

Exploratory Data Analysis (EDA) serves as a foundational milestone in data science workflows, enabling practitioners to evaluate structured data, discover latent patterns, identify distributional traits, and test underlying hypotheses prior to predictive modeling. This report presents an empirical analysis performed on a customer shopping dataset comprising **2,004 observations and 11 features** during Task 3 of the Thiranex Data Science Internship.

Using Python and its data analytics ecosystem (Pandas, NumPy, Matplotlib, and Seaborn), standard quantitative summaries and categorical aggregations were executed. The primary target metric evaluated was transaction spend (Purchase Amount). Key highlights indicate an overall average purchase amount of \$52.58. Subscription status demonstrated the single most notable variation in purchase magnitude (\$54.27 for active subscribers vs. \$46.70 for non-subscribers). Conversely, demographic age metrics demonstrated virtually zero linear correlation ($r = -0.0028$) with purchase amounts.

2. PROJECT OBJECTIVES

Primary Goal: Perform a robust statistical and visual examination of customer transaction characteristics using Python to uncover spending drivers and demographic patterns without making unverified causal assumptions.

- Inspect dataset structural integrity, verifying column data types and missing value counts.
- Compute core descriptive statistical metrics (mean, median, standard deviation, range).
- Perform grouped categorical aggregations across item categories, shipping modes, seasons, and professions.
- Formulate a comprehensive 4x4 visual composite grid (16 plots) capturing univariate and bivariate trends.
- Quantify numerical linear associations via Pearson correlation metrics.

3. DATASET OVERVIEW & DATA INTEGRITY

The analysis was conducted on a verified primary dataset containing 2,004 individual purchase records. Initial preprocessing confirmed that the dataset contained **0 missing or null values** across all 11 attributes.

ATTRIBUTE NAME	DATA TYPE	DESCRIPTION / CONSTRAINTS
CustomerID	Integer	Unique identifier for transaction record
Gender	Categorical	Customer gender identity (Female, Male)
Age	Integer	Customer age in years (Range: 18 – 74)
Items Purchased	Categorical	Specific product name (12 distinct items)
Category	Categorical	Broader merchandise category (4 categories)
Purchase Amount	Continuous	Dollar value of purchase (Range: \$12.00 – \$100.00)
Shipping Type	Categorical	Delivery method (4 distinct options)

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4. NUMERICAL STATISTICAL SUMMARY

Central tendency and dispersion metrics were calculated for continuous variables (Age and Purchase Amount). Both variables exhibit well-bounded parameter distributions.

29.42 yrs MEAN AGE	22.00 yrs MEDIAN AGE	\$52.58 MEAN PURCHASE	\$50.00 MEDIAN PURCHASE
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Metric Parameter	Customer Age (Years)	Purchase Amount (\$)
Mean	29.422655	52.580339
Median	22.000000	50.000000
Standard Deviation	13.050978	29.340990
Minimum Value	18.000000	12.000000
Maximum Value	74.000000	100.000000

Statistical Interpretation: Age exhibits a lower median (22.0) compared to its mean (29.42), reflecting a right-skewed distribution where young adults represent the densest customer segment. Purchase Amount is balanced between \$12.00 and \$100.00 with a standard deviation of \$29.34.

5. GROUPED CATEGORICAL SPENDING ANALYSIS

Average Purchase by Product Category

Category	Avg Spend (\$)
Footwear	53.217125
Clothing	52.841503
Outerwear	52.347458
Accessories	50.400922

Average Purchase by Subscription

Subscription Status	Avg Spend (\$)
Subscribed (Yes)	54.274422
Unsubscribed (No)	46.696429

Average Purchase by Item Type

Item Type	Avg Spend (\$)
Sandals	57.807692
Jeans	53.224880
Blouse	52.873418
Sweater	52.754098
Coat	52.472340

Average Purchase by Season

Season	Avg Spend (\$)
Fall	53.053333
Winter	52.670227
Summer	52.518053
Spring	52.464088

Average Purchase by Gender

Gender	Avg Spend (\$)
Female	53.010924
Male	51.950860

Average Purchase by Profession

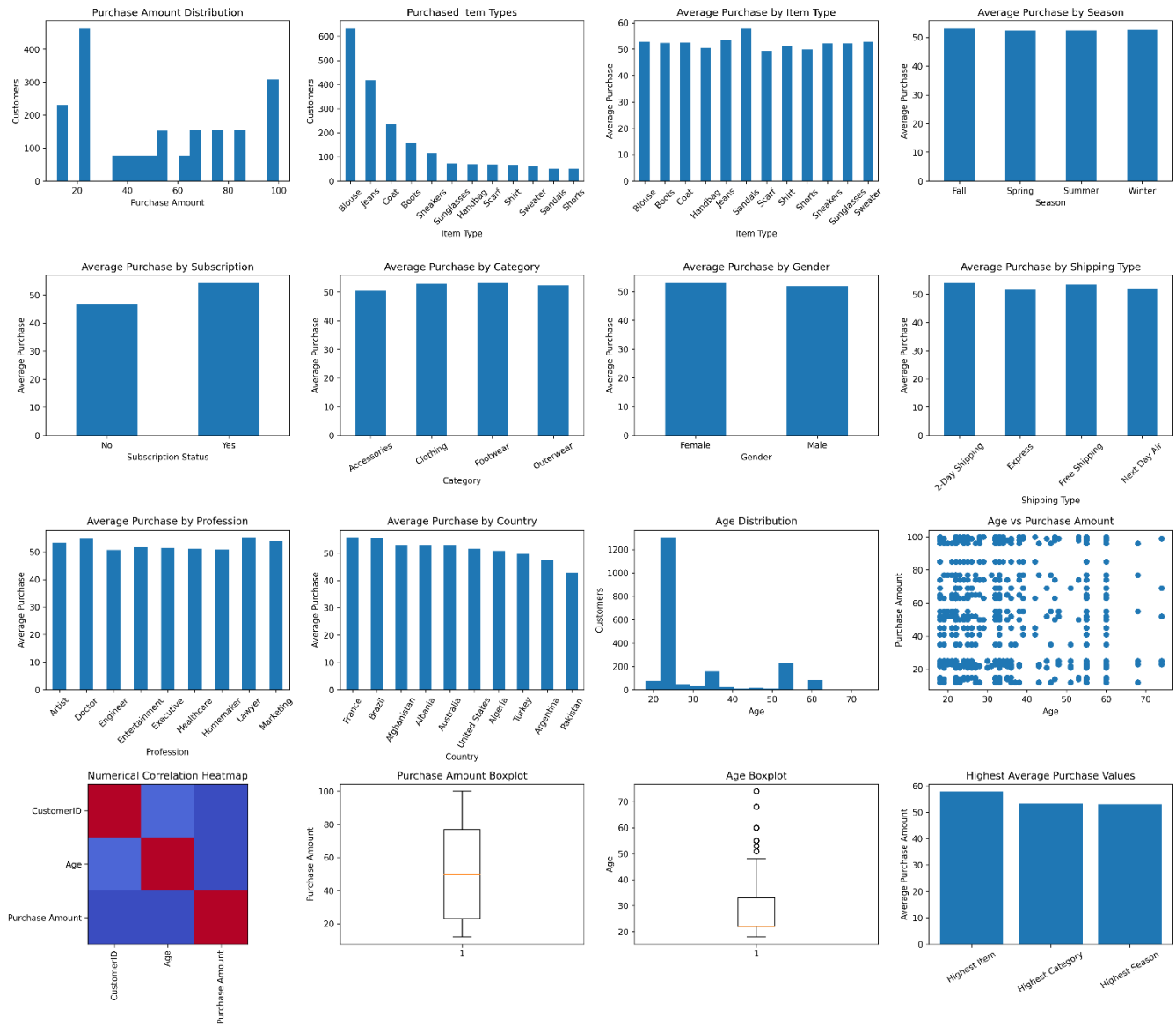
Profession	Avg Spend (\$)
Lawyer	55.267606
Doctor	54.677019
Marketing	53.882353
Artist	53.297381
Entertainment	51.673729

Boots	52.343750	Executive	51.477124
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6. VISUALIZATION ARCHITECTURE (4x4 MATRIX)

To evaluate univariate distributions and multi-variable interactions, 16 individual figures were constructed and arranged into a 4x4 dashboard panel using Matplotlib and Seaborn.

Customer Shopping Dataset - Exploratory Data Analysis



7. CORRELATION ANALYSIS

Pearson correlation coefficients (r) were computed across numerical variables to test for linear dependency.

VARIABLE PAIR	PEARSON COEFFICIENT (r)	ANALYTICAL INTERPRETATION
Age vs. Purchase Amount	-0.002817	Almost zero linear relationship; age does not dictate spend amount.
CustomerID vs. Age	0.059782	Negligible association (expected for index identifier).

CustomerID vs. Purchase Amount	0.000718	Zero linear association (index variable baseline).
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8. KEY ANALYTICAL FINDINGS

- **Subscription Engagement:** Active subscription status showed the most noticeable difference in average purchase amount (\$54.27 vs. \$46.70), reflecting a +\$7.57 delta for subscribers.
- **Product Specifics:** Among individual item types, *Sandals* recorded the highest mean purchase value (\$57.81), whereas *Scarf* recorded the lowest (\$49.32).
- **Category Dominance:** *Footwear* yielded the highest average spending among categories (\$53.22).
- **Seasonal Stability:** *Fall* yielded the highest seasonal average (\$53.05), though differences across all four seasons remained under \$0.60.
- **Linear Independence:** Age showed virtually no linear relationship with purchase magnitude ($r = -0.0028$).

9. CONCLUSION & LEARNING OUTCOMES

This exploratory analysis successfully evaluated 2,004 customer transactions without making unsupported causal claims. The findings suggest that active customer engagement (subscriptions) aligns with higher purchase amounts more strongly than standard age demographics. Through Task 3 of the Thiranex internship, proficiency was deepened in Pandas data aggregation, multi-panel visual plotting using Seaborn, and quantitative reporting using academic standards.